

SoundAround: An Interactive System for High-Resolution, Real-Time Sound Localization

Jeremy Zhengqi Huang
University of Michigan
Ann Arbor, MI, USA
zjhuang@umich.edu

Johnny Qin
University of Michigan
Ann Arbor, MI, USA
leqin@umich.edu

ABSTRACT

Sound is one of the most important cues to be aware of the state of the environment. However, deaf and hard of hearing (DHH) people may have challenges accessing and locating sounds, which hinders their ability to effectively perform everyday tasks like noticing sirens or fire alarms. While technologies like hearing aids and cochlear implants have helped DHH people overcome the limitations regarding sound awareness, they are not affordable, and few are dedicated to assisting users to localize sounds. In this paper, we present SoundAround, an interactive system that helps DHH people interactively identify sound directions using peripheral visual cues. SoundAround indicates sound direction by converting sound signal to sound direction data and mapping the data to an array of LED beads. Our technical evaluations demonstrate that SoundAround can help DHH users localize sound both accurately and responsively in a low-noise environment. However, the localization system may get distracted by loud noises, potentially making users spend more time interpreting the LED signals and identifying the sound direction. Based on the findings, we propose future directions such as implementing adaptive systems to recognize ambient sounds and other noises and prioritize sounds based on the contexts.

INTRODUCTION

People who are deaf and hard-of-hearing (DHH) have limited access to the sounds in their environment, potentially hindering them from effectively performing everyday tasks or being aware of their environment (e.g., noticing sirens and knowing the phone ringing). While technologies like hearing aids have been designed to help DHH people access sound, their ability to help users localize sound is limited [1]. Prior work used desktop-based positional sound visualization to inform DHH users of sound location, but these systems rely on detailed information of the location (e.g., floor plans) and are not portable. Goodman [2] explored using fixed LED on eyeglasses to indicate the direction. While this study demonstrated the value of helping users be aware of sounds through visualizations, the system reported sounds in a limited range (front angles) and resolution (left or right).

Our system, SoundAround, extends on Goodman’s approach [2] by enabling higher-resolution LED signals (15 degrees) and reporting sounds from all directions (360-degree sound reporting). SoundAround is a wearable sound localization device that actively detects sound from the environment and converts sound information to LED signals by mapping the

sound directions to a programmable LED strip with 12 beads. We place the LED signals in users’ peripheral vision to minimize the system’s distraction. With the LED signals as dynamic visual cues, users should be able to interactively identify and track sound directions in real time.

We examined the SoundAround system by conducting three technical evaluations. The first two studies took place in a quiet lab, and the third study took place outdoor and about 150 feet away from a construction site. The results indicated our system’s ability to help users accurately and responsively identify sound directions in a quiet and moderately noisy environment (0-50 dB). The system performance declined when situated in environments with inconsistent or extremely high noise (e.g., construction sites; around 70 dB).

In summary, the SoundAround system offers a high-resolution solution for DHH users to interactively localize sound from all directions.

RELATED WORK

Prior research in sound awareness studied numerous designs for visualizing sound information, mainly to help DHH people identify sound identity [3,4], location [2,3,5], volume, and frequency [5]. Matthews et al. [5] built several basic designs for the desktop screens to visualize sound characteristics (e.g., showing the sound volume with rings). Jain et al. [3] leveraged similar approach using tablets as “smart home” devices to visualize sound identity and location in different locations at home. While these designs provided DHH users with diverse sound information, its stationary, single-context nature limited its usage scenarios. More recent work employed mobile approaches to sound visualization. For example, Goodman et al. [2] designed a smartwatch-based interface where sound direction and loudness were represented by egocentric arcs along with sound identity. While this approach provided glanceable visualizations of sound information, the sound localization may not be accurate due to its wrist-worn nature: human arms move frequently and do not always represent the direction people face. Our approach is inspired by Gorman’s LED-based visualization [3] in that it uses a programmable LED-strip with 12 beads provide DHH users with high-resolution, location-independent, and minimal-effort sound awareness solution. That is, users will be able to perceive the sound direction information without interrupting their current tasks (e.g., raising wrists or taking the phone out of the pocket). However, our system is not able to visualize

information about sound identity and loudness. In addition, despite our effort to minimize distraction from the LED signals, the long-term exposure to LED lights can still cause eye straining.

METHOD

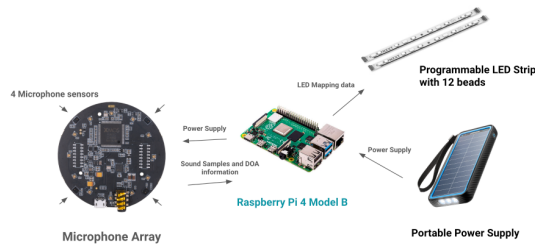


Figure 1. The block diagram of the components of the SoundAround system.

System Overview

The SoundAround system has four components (see Figure 1): 1) a four-microphone array, 2) programmable LED strip, 3) Raspberry Pi, and 4) portable power bank.

We use a four-microphone array from ReSpeaker¹ to detect sounds. The microphone array is placed on top of the helmet to accurately collect sound direction information. We surround the top part of the peripheral visual field with a 12-bead programmable LED strip, with each bead represents an interval of 15 degree of visual field (see Figure 2). We use a Raspberry Pi computer to receive sound information from the microphone array and program the LED beads. We power all the system components with a 10,000mAh power bank.

User Interface

Our user interface consists of 12 LED beads. When there is no sound detected, all 12 beads are off. When there is sound coming from the front, the light bead that points to the direction of the sound source will shine red. If the sound comes from behind, the light beads that points to the “mirrored” direction of the sound source will shine green (see Figure 2). The lighting patterns will also adapt to user movements. For example, if users tilt their head while the sound is still present, the LED signals will recalibrate by lighting up a different bead according to the new direction user is facing towards.

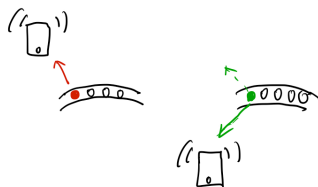


Figure 2. Sketch of the LED-based user interface.

Sound Localization and Visualization

Sound is sampled from the four microphone-array at 16,000Hz. The system processes the sound by adopting a TDOA (Time Difference of Arrival) localization scheme that measures the time difference for the sound to reach each microphone and uses it to calculate the direction of arrival (DOA). We set the DOA output to 10Hz to strike a balance between power consumption and responsive real-time localization.

Note: In this paper, we adopt this scheme for counting degrees: straight front will be 0 degree; the degree value then implements clockwise. For example, 90-degree angle means straight to the right, while 270-degree angle means straight to the left.

We then map the DOA output, which is a continuous stream of numerical directional value ranging from 0 to 360 degrees to the 12 programmable LED beads, so that every individual bead represents a 15-degree interval (7.5 degree from the left + 7.5 degree to the right). For example, the bead at the middle represents 0 degree, but will light up when the sound comes from 352.5- to 7.5-degree angle. We created a lightweight Python program that runs sound localization and LED control programs simultaneously using a shared memory.

To make sure that the beads will dim off after 1 second of no sound activity, the system will buffer the most recent 10 DOA data (one second) and, if all data share the same value, dims the beads.

At the later stage of the project, we programmed the parameters on microphone array to enable non-stationary noise suppression.

TECHNICAL EVALUATION AND USER STUDY

We conducted three technical evaluations to test SoundAround’s ability to help DHH users identify sound direction *accurately* and *responsively* in contexts with different levels of background noises present.

Participants

We recruited 5 participants. We asked all participants to wear a noise-cancellation headphone to limit their access to the environment sound. We want to stress that this approach is not equivalent to the deaf and hard of hearing experience.

Study 1

The purpose of the Study 1 to test the system’s ability to help users identify sound direction using solely the LED signals. Participant sat in a quiet lab environment on one side of the table and wore the SoundAround system. Researchers stood on the other side of the table simulating the phone ringing by playing the ringtones aloud. Participants could not see the phone or researcher as there was a tall, solid cardboard standing in between (see Figure 3). The boards were segmented into 5 equal-length sections and were labeled D1

¹ <https://www.seedstudio.com/ReSpeaker-Mic-Array-v2-0.html>

(“Direction 1”) to D5 (“Direction 5”). During the experiment, researchers played the ringtone behind each *Direction*, and participants were told to say aloud which *Direction* the sound source was at. Each participant will have 5 trials, with each trial corresponding to one *Direction*.



Figure 3. Study 1 setup.

Data

For each trial, we noted participants’ reported sound direction and measured the error distance from participants’ pointed direction to the correct direction. For example, if the phone rings at D5 and participants said “3”, the error distance will be 2. We also counted the correct identifications of the sound direction.

Study 2

The purpose of this study is to test the system’s ability to help users identify sound direction using solely the LED signals, but this time for sounds that are behind the users. This study simulates an office setting where sounds like fire alarms take place are out of DHH people’s sights. During the study, participants sit in front of a study desk while facing the wall, and researchers played sounds at 180-, 225-, and 270-degree direction (see Figure 4). We only tested three directions due to the space limitation. Participants were told to point at the perceived direction of the sound source.

Data

For each trial, we recorded the participants’ response as “correct” if the participants pointed at the exact direction of the sound source or a direction that is very close to that of the sound source. We recorded the response as “incorrect” otherwise.

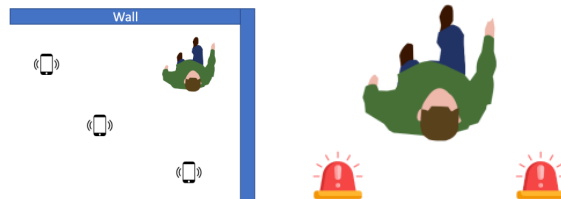


Figure 4 (L). Study 2 setup. Each phone ringing icon represents a trial.

Figure 5 (R). Study 3 setup.

Study 3

The aim of Study 3 is to evaluate the system’s ability to inform DHH users of sound from behind with various levels of background noise present. We simulated a scenario where DHH people walked on the street with an emergency vehicle and siren approaching from behind. Researchers played a siren sound with increasing volume (0 to 96 dB) while standing 5 feet behind the participants on the left and right side (see Figure 5). Participants were told to raise left hand if they thought the siren sound was “approaching” from the left and vice versa. There were 4 trials (left, right, left, right) for each participant (20 trials in total). For the last two trials, we asked participants to perform the same task without the noise cancellation headphone. We consider those two trials for baseline.

After completing the first batch, we discovered the on-device noise suppression feature. We then enabled this feature and conducted two additional tests with P2 and P5 (labeled as P2N and P5N).

Study 3 Data

For each trial, we measured the reaction time, the time participants took to raise their hand after researchers began playing the siren sound, and the labeled the response as “correct” (correct direction indicated) or “incorrect” (incorrect direction indicated).

RESULTS

We present the results from three technical evaluations.

Study 1 Results

The detailed results are listed in Table 1. Overall, the success rate across participants and trials is 88%. Our system performed the best at helping participants identify sound direction accurately at D2 (324-degree) and D3 (0-degree). Even when there were errors, participants’ perceived directions were generally close to the correct ones.

	D2	D4	D1	D3	D5
P1	0	0	1	0	0
P2	0	0	0	0	0
P3	0	0	0	0	0
P4	0	0	0	0	1
P5	0	1	0	0	0
Avg.	0	0.2	0.2	0	0.2
Acc. %	100	80	80	100	80

Table 1. Study 1 results. “Avg.” means average error distance, and “Acc. %” means accuracy percentage for each direction.

Study 2 Results

The detailed results of Study 2 are listed in Table 2. Overall, the accuracy for pointing out sound directions across all trials is 93.3%. Together with results from Study 1, these results demonstrated our system’s ability to accurately localize sounds from all directions.

	T1 (180)	T2 (225)	T3 (270)
P1	CORRECT	CORRECT	CORRECT
P2	CORRECT	CORRECT	CORRECT
P3	INCORRECT	CORRECT	CORRECT
P4	CORRECT	CORRECT	CORRECT
P5	CORRECT	CORRECT	CORRECT
Acc. %	80%	100%	100%

Table 2. Study 2 results. “Acc. %” means accuracy percentage for each direction.

Study 3 Results

The detailed results of Study 3 are listed in Table 3 and Figure 4. Overall, participants were 23% more accurate than the baselines at identifying siren sound directions with the help of SoundAround system. The system performed very well with low to moderate background noise, but the accuracy declined as the noise level went up (e.g., 50% at 67 dB and 72 dB). Still, users’ average accuracy in identifying sound direction with the system was the same as the baseline accuracy with loud background noise, indicating SoundAround’s potentials to provide equitable sound access to DHH people. After we enabled the noise suppression feature, the accuracy went up to at least 75% for both P2 and P5 with moderately high noise level (i.e., 59 and 63 dB).

	T1 (LEFT)	T2 (RIGHT)	T3 (LEFT-B)	T4 (RIGHT-B)	Acc. %
P1 (37dB)	CORRECT	CORRECT	CORRECT	CORRECT	100%
P2 (42dB)	CORRECT	CORRECT	CORRECT	INCORRECT	75%
P3 (67dB)	CORRECT	INCORRECT	INCORRECT	CORRECT	50%
P4 (72dB)	INCORRECT	CORRECT	INCORRECT	CORRECT	50%
P5 (52dB)	CORRECT	CORRECT	CORRECT	CORRECT	100%
P2N (59dB)	CORRECT	CORRECT	CORRECT	CORRECT	100%
P5N (63dB)	CORRECT	CORRECT	CORRECT	INCORRECT	75%
Acc. %	80%	80%	50%	80%	
Acc. % (N)	100%	100%	100%	50%	

Table 3. Study 3 results for accuracy. T3 and T4 are trials for baselines. P2N and P5N represent two additional tests after we enabled the noise suppression. “Acc. %” means accuracy percentage for each trial. “Acc. % (N)” means the accuracy percentage for each trial after turning on the noise suppression.

When the noise level is low, participants were able to quickly notice the siren sound using SoundAround, sometimes with shorter reaction time than baselines (37 dB). However, the reaction time became significantly longer when the noise level went up. Moreover, compared to the baselines, participants took much longer to indicate sound direction when using SoundAround system, regardless of the presence of noise suppression. One possible explanation is that the irregular noise from the construction site distracted the microphones, as P2 and P4 reported that the LED signals kept “switching back and forth”.

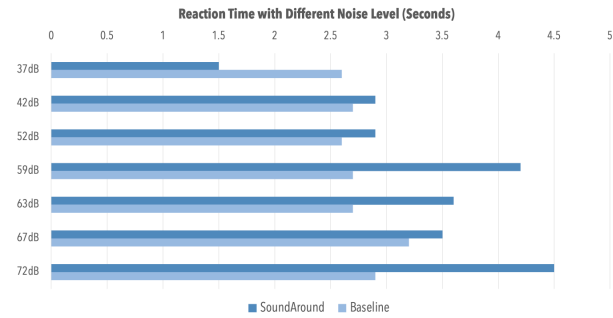


Figure 6. Study 3 results: user reaction time to the siren with different noise levels.

DISCUSSION AND CONCLUSION

In the present study, we developed a wearable interactive system that used LED signals as visual cues to help DHH people localize sound. The technical evaluations shows that our system can help users accurately and responsively identify sound directions in low noise environment. While our system is still fairly accurate for localizing sounds when the noise level goes up, it is more likely to be distracted by the noise, making it harder for DHH users to identify sound directions using the LED signals. This indicates the need for the system to adapt to different environments by tuning out the ambient sounds or background noise and focusing on more critical sounds. For example, to achieve context awareness, our system can employ ambient sound recognition tools [7] to identify the context and prioritize sounds that are more likely to take place.

We have also acquired ideas for future improvements from instructors’ feedback. SoundAround can also incorporate a LED-based sound recognition program that prompts the beads to have distinctive lighting patterns for different sound categories. For example, if the sounds are about critical events (e.g., fire alarms), the beads can blink red to show urgency.

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